AMATH/PMATH 331: Applied Real Analysis Fall 2026

Assignment 1 Submission; due Thursday, September 24, 2026

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[1] Let $E \subseteq \mathbb{R}$ be nonempty. We say that $t \in \mathbb{R}$ is a *lower bound* of E if $t \leq x$ for all $x \in E$. We say that E is *bounded below* if there exists a lower bound of E. We say that $t \in \mathbb{R}$ is a *greatest lower bound*, or *infimum*, of E if it is a lower bound of E and if $t \geq s$ for any lower bound s of E. If E has an infimum, it is denoted $\inf E$.

[a] Let E be a nonempty subset of $\mathbb{R}$ which is bounded below. Find a way to use the completeness axiom to prove that $\inf E$ exists. *Hint:* Consider the subset $-E = \{-x : x \in E\}$.

Answer:

Proof. Since E is nonempty and bounded below, there exists $\alpha \in \mathbb{R}$ such that

$$\forall x \in E : \ \alpha \leq x$$

Consequently,

$$\forall x \in E : \ -\alpha \geq -x$$

.

Define

$$-E = \{-x : x \in E\}.$$

Then, $-\alpha$ is an upper bound of $-E$.

By the completeness axiom, $\sup(-E)$ exists. Let

$$\beta = \sup(-E).$$

It follows that $\forall x \in -E : \ \beta \geq x$. And therefore, $\forall x \in E : \ -\beta \leq x$. So $-\beta$ is a lower bound of E.

Next, take any other lower bound of E, call it s. From the same logic as above, it follows that $-s$ is an upper bound of $-E$. Since β is a supremum of $-E$, it follows that $\beta \leq -s$.

So, $-\beta \geq s$. Therefore, since $-\beta$ is a lower bound of E and it is at least as large as any other lower bound, it follows that $-\beta = \inf E$ □

[b] Let $\varnothing \neq A \subseteq B \subseteq \mathbb{R}$. Suppose B is bounded below (so $\inf B$ exists). Prove that A is bounded below (so $\inf A$ exists) and that $\inf A \geq \inf B$.

Answer:

Proof. If B is bounded below, then $\exists \alpha : \ \forall x \in B : \ \alpha \leq x$.

Since $A \subseteq B$, we know that $\forall y \in A : \ y \in B$.

Consequently,

$$\forall y \in A : \alpha \leq y$$

Therefore, α is a also a lower bound of A. And from 1[a], we can conclude that $\inf A$ also exists.

Now, since A is a subset of B and $\inf B$ is a lower bound of B, then $\inf B$ must also be a lower bound of A. Since $\inf A$ must be greater then or equal to any other lower bounds of A, we have that

$$\inf A \geq \inf B$$

□

[c] Let A, B be nonempty subsets of $\mathbb{R}$, both bounded below (so both $\inf A$ and $\inf B$ exist). Define

$$A + B = \{a + b : a \in A,\ b \in B\}.$$

Prove that $A + B$ is bounded below and that $\inf(A + B) = \inf A + \inf B$.

Answer:

Proof.

$$\begin{aligned} \forall a \in A :\ & \inf A \leq a \\ \forall b \in B :\ & \inf B \leq b \\ \therefore \forall a \in A, b \in B :\ & \inf A + \inf B \leq a + b \\ \therefore \forall x \in A + B :\ & \inf A + \inf B \leq x \end{aligned}$$

So $\inf A + \inf B$ is a lower bound of $A + B$. Consequently, $\inf(A + B)$ exists.
Now, we know that

$$\begin{aligned} \forall \varepsilon > 0 :\ & \exists \alpha \in A :\ \inf A + \frac{\varepsilon}{2} > \alpha \\ \forall \varepsilon > 0 :\ & \exists \beta \in B :\ \inf B + \frac{\varepsilon}{2} > \beta \end{aligned}$$

Therefore,

$$\forall \varepsilon > 0 :\ \exists \alpha \in A, \beta \in B :\ \inf A + \inf B + \varepsilon > \alpha + \beta$$

This means that no number greater then $\inf A + \inf B$ can be a lower bound of $A + B$. Therefore,

$$\inf(A + B) = \inf A + \inf B$$

□

[2] Let x, y be two real numbers such that $y - x > 1$. (In particular, $x < y$.)

[a] If x is positive, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.

Answer:

Proof. By the Archimedean property, the set $E = \{n \in \mathbb{N} : \ n > x\} \subset \mathbb{N} \subset \mathbb{Z}$ is not empty.
x is a lower bound of E, so by the completeness axiom, $\inf E$ exists. And since $E \subset \mathbb{Z}$, $\inf E \in E$. Let $\inf E = k$.
Then $x < k$. Now, assume for the sake of contradiction that $x < k - 1$. Since $k - 1 \in \mathbb{Z}$, then $k - 1 \in E$. But $k - 1 < k$ and $k = \inf E$, which is a contradiction.
Therefore, $k - 1 \leq x$. $k \leq x + 1 < y$.
So, $x < k < y$.

□

[b] If x is negative and y is positive, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.

Answer:

Proof. Take $k = 0 \in \mathbb{Z}$. □

[c] If y is negative, prove that there exists an integer $k \in \mathbb{Z}$ such that $x < k < y$.

Answer:

Proof. Since $y < 0$, we have $-y > 0$. Moreover,

$$(-x) - (-y) = y - x > 1$$

So applying 2[a] to $-x$ & $-y$, we get that $\exists n \in \mathbb{Z} : \ -y < n < -x$.

$$x < -n < y$$

Taking $k = -n \in \mathbb{Z}$ proves the result. □

[3] The parts of this problem are *not related*.

[a] Let $x \in \mathbb{R}$. Consider the set $E = \{q \in \mathbb{Q} : q < x\} \subset \mathbb{Q} \subset \mathbb{R}$. Show that E is nonempty and bounded above, so $\sup E$ exists. Show also that $\sup E = x$.

Answer:

Proof. Applying the Archimedean property to $-x$ we get that $\exists n \in \mathbb{N} :\ n > -x$. So, $x > -n$. Therefore, $-n \in E$, so E is non-empty.

Also, since x is strictly larger than any element of E, x is a lower bound of E.

Since E is bounded above and non-empty, $\sup E$ exists.

Let $\sup E = \alpha$. We know that $\alpha \leq x$. Now, assume for the sake of contradiction, that $\alpha < x$.

One of the propositions proved in class showed that between any two distinct real numbers, there exists a rational number. That is:

$$\exists q \in \mathbb{Q} :\ \alpha < q < x$$

But if q is rational and smaller then x, then it should be a part of E. But this contradicts α being an infimum of E. Therefore, our assumption must be false and

$$\sup E = x$$

□

[b] Let $x, y \in \mathbb{R}$ with $x < y$. Prove that there exists an *irrational* (not rational) number z such that $x < z < y$. *Hint:* You may use the fact that $\sqrt{2}$ is irrational.

Answer:

Proof. We will take the interval (x, y) and shift it by $\sqrt{2}$.

$$x - \sqrt{2} < y - \sqrt{2}$$

By the proposition shown in class, we know that

$$\exists q \in \mathbb{Q} :\ x - \sqrt{2} < q < y - \sqrt{2}$$

Therefore, $x < q + \sqrt{2} < y$. Since we know that $\sqrt{2}$ is irrational, then $z = q + \sqrt{2}$ is also irrational. □

[4] Let $C > 0$ be a fixed positive number.

[a] Let $x \geq 0$, and suppose that for any $\varepsilon > 0$, we have $x < C\varepsilon$. Show that for any $\eta > 0$, we have $x < \eta$.

Answer:

Proof. Let $\eta > 0$ and choose

$$\varepsilon = \frac{\eta}{C}$$

Then we know that $x < C\varepsilon = C\frac{\eta}{C}$. Therefore, $x < \eta$ □

[b] In the situation of part [a], show that we must have $x = 0$. *Hint:* Find a way to use the Archimedean property.

Answer:

Proof. Suppose for the sake of contradiction that $x > 0$.

By the Archimedean property, we get that

$$\exists n \in \mathbb{N} : \ n > \frac{1}{x}$$

Therefore, $x > \frac{1}{n}$. But we can take $\eta = \frac{1}{n} > 0$ in 4[a] and get that $x < \frac{1}{n}$. This is a contradiction, so our assumption must be wrong. Therefore, $x = 0$. □

[5] In this problem you will give an alternative proof of the Cauchy–Schwarz inequality. The inequality says that

$$-\|x\|\|y\| \leq \langle x, y\rangle \leq \|x\|\|y\|.$$

Both inequalities are trivially seen to be equalities if either $x = 0$ or $y = 0$, so we suppose that both vectors are nonzero.

By consider the norm of the vector

$$z = \frac{x}{\|x\|} \pm \frac{y}{\|y\|},$$

Show that:

1. both inequalities are true.
2. the first inequality becomes an equality if and only if $y = tx$ for some $t < 0$.
3. the second inequality becomes an equality if and only if $y = tx$ for some $t > 0$.

Answer:

Proof. Let $u = \frac{x}{\|x\|}$, $v = \frac{y}{\|y\|}$ and $z = u \pm v$.

$$\|u\| = \left\|\frac{x}{\|x\|}\right\| = \frac{\|x\|}{\|x\|} = 1 \quad \|v\| = \left\|\frac{y}{\|y\|}\right\| = \frac{\|y\|}{\|y\|} = 1$$
$$\|z\|^2 = \|u \pm v\|^2 = \langle u \pm v, u \pm v\rangle = \|u\|^2 + \|v\|^2 \pm 2\langle u, v\rangle = 2 \pm 2\langle u, v\rangle$$
$$\|z\|^2 = 2(1 \pm \langle u, v\rangle) \geq 0 \implies -1 \leq \langle u, v\rangle \leq 1$$
$$\langle u, v\rangle = \left\langle \frac{x}{\|x\|}, \frac{y}{\|y\|} \right\rangle = \frac{\langle x, y\rangle}{\|x\|\|y\|}$$
$$\therefore -1 \leq \frac{\langle x, y\rangle}{\|x\|\|y\|} \leq 1 \implies -\|x\|\|y\| \leq \langle x, y\rangle \leq \|x\|\|y\|$$

Next, for the equality conditions, we will first prove

$$y = tx \text{ for some } t > 0 \iff \langle x, y\rangle = \|x\|\|y\|$$

First, suppose $y = tx$ for some $t > 0$. We have to prove that $\langle x, y\rangle = \|x\|\|y\|$.

$$\langle x, y\rangle = \langle x, tx\rangle = t\langle x, x\rangle = t\|x\|^2$$
$$\|y\| = \|tx\| = t\|x\|$$
$$\therefore \langle x, y\rangle = \|x\|\|y\|$$

Secondly, suppose $\langle x, y\rangle = \|x\|\|y\|$. We have to prove that $y = tx$ for some $t > 0$. Since $\langle x, y\rangle = \|x\|\|y\|$, we have that $\langle u, v\rangle = \frac{\langle x, y\rangle}{\|x\|\|y\|} = 1$. Let $z = u - v$.

$$\|z\|^2 = 2 - 2\langle u, v\rangle = 0 \implies u = v$$
$$\frac{x}{\|x\|} = \frac{y}{\|y\|} \implies y = x\frac{\|y\|}{\|x\|} \qquad \|y\| > 0, \quad \|x\| > 0$$
$$\therefore y = tx \qquad \text{For } t = \frac{\|y\|}{\|x\|} > 0$$

This concludes the proof that $y = tx$ for some $t > 0 \iff \langle x, y\rangle = \|x\|\|y\|$

Now we have to prove

$$y = tx \text{ for some } t < 0 \iff \langle x, y\rangle = -\|x\|\|y\|$$

Consider the vector $x' = -x$ and apply the proof above to show that

$$y = t'x' \text{ for some } t' > 0 \iff \langle x', y\rangle = \|x'\|\|y\|$$

Now,

$$\begin{aligned}
&y = t'(-x) \text{ for some } t' > 0 \iff \langle -x, y\rangle = \|-x\|\|y\| \\
&y = (-t')x \text{ for some } t' > 0 \iff -\langle x, y\rangle = \|x\|\|y\| \qquad \|x\| = \|-x\| \\
&y = (-t')x \text{ for some } t' > 0 \iff \langle x, y\rangle = -\|x\|\|y\| \\
&\text{Take } t = -t' < 0 \\
&y = tx \text{ for some } t < 0 \iff \langle x, y\rangle = -\|x\|\|y\|
\end{aligned}$$

□

[6] Let

$$E = \overline{B_r(x_0)} = \{x \in \mathbb{R}^n : \|x - x_0\| \leq r\}$$

be a closed ball. Prove that E is *not open*.

Answer:

Proof. We have to prove that $\exists x \in E : \ \forall \delta > 0 : \ B_\delta(x) \not\subseteq E$.
Equivalently: $\exists x \in E : \ \forall \delta > 0 : \exists y \notin E : \ \|y - x\| < \delta$.

Take any x such that $\|x - x_0\| = r$. Then, for any $\delta > 0$ construct the vector:

$$y = x + \frac{\delta}{2r}(x - x_0)$$

$$\|y - x_0\| = \left\|x + \frac{\delta}{2r}(x - x_0) - x_0\right\| = \left\|(x - x_0)(1 + \frac{\delta}{2r})\right\|$$
$$\|y - x_0\| = (1 + \frac{\delta}{2r})\|x - x_0\| = (1 + \frac{\delta}{2r})r = r + \frac{\delta}{2}$$
$$\|y - x_0\| = r + \frac{\delta}{2} > r$$

Therefore, $y \notin E$.

$$\|y - x\| = \left\|x + \frac{\delta}{2r}(x - x_0) - x\right\| = \left\|\frac{\delta}{2r}(x - x_0)\right\|$$
$$\|y - x\| = \frac{\delta\|x - x_0\|}{2r} = \frac{\delta r}{2r} = \frac{\delta}{2}$$
$$\therefore \|y - x\| = \frac{\delta}{2} < \delta$$

Thus, we have proved that $\exists x \in E : \ \forall \delta > 0 : \exists y \notin E : \ \|y - x\| < \delta$, which is equivalent to saying E is not open. □

[7] Let $F = \{x_1, \ldots, x_N\}$ be a finite subset of $\mathbb{R}^n$. Prove that F is *closed*.

Answer:

Proof. We have to prove that $\forall x \in F^c : \exists \delta > 0 : B_\delta(x) \subseteq F^c$.
Let $y \in F^c$. So $\forall x_i \in F : y \neq x_i$. Therefore, $\forall x_i \in F : \|y - x_i\| > 0$
Since F is a finite set, the set $M = \{\|y - x_i\| : i \in [N]\}$ is also finite, so $min(M)$ exists.
Take $\delta = min(M) = min(\|y - x_i\|) > 0$.
Let $z \in B_\delta(y) \implies \|z - y\| < \delta$.
By the triangle inequality: $\forall x_i \in F : \|z - x_i\| \geq \|y - x_i\| - \|y - z\|$.
Now, since $\delta = min(\|y - x_i\|)$, we know that $\forall x_i \in F : \delta \leq \|y - x_i\|$
So, the inequality becomes: $\forall x_i \in F : \|z - x_i\| \geq \delta - \|y - z\|$.
But we know that $\|y - z\| < \delta$. Therefore, $\forall x_i \in F : \|z - x_i\| > 0$.
So, $\forall x_i \in F : z \neq x_i \implies z \in F^c \implies B_\delta(y) \subseteq F^c$.
Therefore F^c is open $\implies$ F is closed. □

[8] Let $a \in \mathbb{R}^n \setminus \{0\}$, and $c \in \mathbb{R}$. Consider the subset

$$E = \{x \in \mathbb{R}^n : \langle x, a\rangle < c\}.$$

Prove that E is open. *Hint:* Observe that

$$|\langle y, a\rangle - \langle z, a\rangle| = |\langle y - z, a\rangle| \leq \|y - z\|\|a\|$$

by Cauchy–Schwarz.

Answer:

Proof. Take arbitrary $y \in E$. Set

$$\delta = \frac{c - \langle y, a\rangle}{\|a\|}.$$

Since $\langle y, a\rangle < c$ and $\|a\| > 0$, we have $\delta > 0$.

Let $z \in B_\delta(y)$. Then $\|z - y\| < \delta$, so by Cauchy–Schwarz,

$$\begin{aligned}
\langle z, a\rangle &= \langle y, a\rangle + \langle z - y, a\rangle \\
&\leq \langle y, a\rangle + |\langle z - y, a\rangle| \\
&\leq \langle y, a\rangle + \|z - y\|\|a\| \\
&< \langle y, a\rangle + \delta\|a\| \\
&= c.
\end{aligned}$$

Thus $z \in E$, and hence $B_\delta(y) \subseteq E$. Since $y \in E$ was arbitrary, E is open. □

[9] Let $C \subset \mathbb{R}^n$ be a non-empty proper *closed* subset of $\mathbb{R}^n$. Let $x \in \mathbb{R}^n \setminus C$, which exists because C is a proper subset. Define the subset $E \subset \mathbb{R}$ by

$$E = \{r > 0 : \ B_r(x) \cap C = \emptyset\}.$$

[a] Explain why $E \neq \emptyset$

Answer:

Proof. Since C is a closed subset of $\mathbb{R}^n$, we have that $C^c = \mathbb{R}^n \setminus C$ is an open subset of $\mathbb{R}^n$. By definition, this means that $\forall x \in C^c : \ \exists r > 0 : \ B_r(x) \subseteq C^c$.
Which is equivalent to saying $\forall x \in \mathbb{R}^n \setminus C : \ \exists r > 0 : \ B_r(x) \cap C = \emptyset$.
Since the given $x \in C^c$, there exists $r > 0$ such that $B_r(x) \cap C = \emptyset$. Hence $r \in E$, so $E \neq \emptyset$. □

[b] Prove that there exists an upper bound for E.

Answer:

Proof. C is non-empty, so consider arbitrary $y \in C$ and look at the distance $\|x - y\|$.
Assume, for the sake of contradiction, that $\exists r \in E : \ r > \|x - y\|$. This would mean that $y \in B_r(x)$. And this implies that $y \in B_r(x) \cap C$, which is a contradiction.
Therefore, $\forall r \in E : \ \|x - y\| \geq r \implies \|x - y\|$ is an upper bound of E. □

[c] Thus, by the completeness axiom, $\sup E$ exists.
Show that $\sup E \leq \|x - y\|$ for all $y \in C$.

Answer:

Proof. In part [b] we proved that $\|x - y\|$ is an upper bound of E for an arbitrary y and so it must hold for all $y \in C$.
Therefore, $\|x - y\|$ is an upper bound of E for all $y \in C$.
And by the definition of the supremum, $\sup E$ is less than or equal to all other upper bounds, so

$$\sup E \leq \|x - y\| \text{ for all } y \in C$$

□

[10] Let $(x_k)_{k\in\mathbb{N}}$ be a sequence in $\mathbb{R}^n$, and suppose that there exists $r \in (0,1)$ and $C > 0$ such that

$$\|x_{k+1} - x_k\| \leq Cr^k$$

Prove that (x_k) is Cauchy.

Hint: Recall that $r^{k+m-1} + r^{k+m-2} + \cdots + r^k = r^k(r^{m-1} + r^{m-2} + \cdots + 1) = r^k \frac{1-r^m}{1-r}$. You may also use without proof the fact (which is easy to show using the Archimedean Property) that: since $r \in (0,1)$, given any $\delta > 0$, there exists $N \in \mathbb{N}$ such that $k \geq N$ implies $r^k < \delta$.

Answer:

Proof. Let arbitrary $\varepsilon > 0$. Let $\delta = \varepsilon \frac{1-r}{C}$. Since $0 < 1 - r < 1$ and $C > 0$, $\delta > 0$. From the hint above, we have that $\exists N \in \mathbb{N} : \; k \geq N \implies r^k < \delta$.

Therefore, $r^k < \varepsilon \frac{1-r}{C}$ for all $k \geq N$.

Now, take arbitrary $k, l \geq N$. Without loss of generality, assume $l \geq k$. Then, $l = k + m$ for some $m \geq 0$.

$$\begin{aligned} x_{k+m} - x_k &= x_{k+m} - x_{k+m-1} + x_{k+m-1} - \cdots - x_{k+1} + x_{k+1} - x_k \\ \|x_{k+m} - x_k\| &= \|(x_{k+m} - x_{k+m-1}) + \cdots + (x_{k+1} - x_k)\| \\ &\leq \|x_{k+m} - x_{k+m-1}\| + \cdots + \|x_{k+1} - x_k\| \\ &\quad \text{(By the triangle inequality)} \end{aligned}$$

Now, we know that $\exists r \in (0,1) : \exists C > 0 : \; \|x_{k+1} - x_k\| \leq Cr^k$ and we can apply this to every norm in the sum above.

$$\begin{aligned} \|x_{k+m} - x_k\| &\leq Cr^{k+m-1} + Cr^{k+m-2} + \ldots Cr^k \\ &\leq C(r^{k+m-1} + r^{k+m-2} + \ldots r^k) \\ &\leq Cr^k \frac{1-r^m}{1-r} \\ &\qquad\qquad 0 < r < 1 \implies 0 < r^m \leq 1 \\ &\qquad\qquad\qquad\quad 0 \leq 1 - r^m < 1 \\ &< r^k \frac{C}{1-r} = \varepsilon \frac{1-r}{C} \frac{C}{1-r} = \varepsilon \end{aligned}$$

Since ε, k, l were arbitrary, it holds that:

$$\forall \varepsilon > 0 : \; \exists N \in \mathbb{N} : \; k, l \geq N \implies \|x_l - x_k\| < \varepsilon$$

Therefore, $(x_k)_{k\in\mathbb{N}}$ is Cauchy. □